Independent Peer Review Report for the Alaska-wide Sablefish Assessment

Meeting at Auke Research Center

Juneau, Alaska

June 16-18, 2026

Prepared for Center of Independent Experts (CIE)

By

Dr. Peter Stephenson

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1. Executive Summary

The Assessment Review for Alaska-wide sablefish was conducted on June 16-18 June 2026 in Juneau, Alaska. Present at the review meeting were the review panel of five, including three CIE reviewers, two presenters, and 3-4 other NOAA participants. Another group of NOAA employees viewed the meeting online.

The background materials were concise, well structured, and readily available before the meeting. The presentations by the Assessment Team were informative and well organized. During the meetings, the Review Panel made suggestions on what could be investigated, and on most occasions, these had already been considered, and the queries were answered immediately.

The assessment models are implemented in the generalized R-TMB based Spatial Populations over Regional Components (SPoRC) stock assessment package. The data for the model runs was readily available enabling the reviewers to run the model before the meeting. The software is appropriate for the assessment. The justification for the preferred assessment model, FAA, is appropriate, and the conclusions drawn are reliable.

Considering the reports supplied and the presentations at the Review Meeting, I conclude:

> The Fleets As Areas (FAA) assessment provided by the Assessment Group is a superior assessment to the single area model, primarily because of its ability to cope with the missing biannual longline survey data in Gulf of Alaska.
>
> Although the FAA assessment had more sparce data inputs and was unstable for some parameter formulations, it is scientifically sound and provided reliable projection outputs for calculation of Tier 3 reference points.
>
> The FAA assessment provided good evidence that the Alaska-wide sablefish stock is not overfished and is not undergoing overfishing.

The Spatial Model, considered as an alternative assessment tool, will be valuable in the future, but is currently unstable due to paucity of age composition and movement information, difficulties in determining suitable selectivity formulations, and confounding of movement with recruitment. The spatial model should be further investigated as it provides a model structure and outputs that are conceptually more understandable than the FAA model. With further investigation and more comprehensive input data, the 5-area spatial model may be a better choice in the future.

2. Background

2.1 Overview

A Review Panel meeting for North Pacific Fishery Management Council on the Alaska-wide sablefish fishery assessment was held at the NOAA Alaska Fisheries Science Center, Auke Bay Laboratories, on June 16 to 19, 2026.

The Review Panel comprised chairperson, Dr Aaron Berger, appointees of the Centre for Independent Experts, Dr Sam Subbey (CIE), Klaas Hartmann (CIE), Mr Peter Stephenson (CIE), and Dr Chris Lumsford. The other attendees of the meeting and their affiliated organizations are shown in Appendix 3.

2.2 Performance Work Statement, Review Report Requirements and Agenda

The Performance Work Statement is shown in Appendix 2. Imbedded in this document were the links to all the documents required for the review. The participants at the meeting are shown in Appendix 3.

The Peer Review Report Requirements provided by the CIE are presented in Annex 1. This required the Review Panel members to participate in an independent peer review of the stock assessment and prepare an independent CIE report of the assessment and the review process. The meeting agenda is shown in Annex 3.

2.3 Acknowledgements

I gratefully acknowledge the contributions of all participants in the review meeting who helped make it both informative and productive. I particularly commend the presenters, Dr Daniel Goethel and Dr Matt Cheng, for the substantial effort invested in the stock assessment and for preparing high-quality supporting documentation. I also thank the Chair, Dr Aaron Berger, for leading a well-organised and efficient meeting, and Dr Chris Lunsford for his assistance with logistics and for helping to make our stay in Juneau enjoyable. It was a pleasure to work with the other CIE reviewers, whose contributions added valuable perspectives to the meeting discussions.

3. Review of Activities

Prior to the meeting, I read the six core documents that were supplied on a GitHub repository (Appendix 1). These were the primary documents essential to understanding the stock assessment, and consisted of a summary document, and five appendices. These documents were well written and concise and enabled me come to the meeting with a good understanding of the sablefish assessment. In addition, there were several supporting documents (Appendix 1), which I perused.

In prepared reports for a review, it is important that units of measurement like “tons” and “metric tonnes” are avoided as this can lead to confusion.

The files used for the assessment model (Appendix 1) were also available prior to the meeting, enabling the CIE reviewers to run the model prior to the meeting.

During the review meeting, the CIE reviewers asked many questions about different model formulations, and in most cases, to their credit, the assessment scientists had already considered them and were able to answer without having to do additional model runs.

Although preparation of a summary report was not required for this review, the meeting Chair determined that such a report would be useful in providing the assessment team with timely feedback and recommendations for future work. A summary report was prepared and is provided in Appendix 4.

4. Findings Related to the Terms of Reference

The terms of reference for the Alaska wide sablefish stock assessment review are shown in Annex 2. This report contains my evaluation of the assessments in terms of the Terms of Reference supplied to the reviewers.

4.1 Terms of Reference 1

Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.

- ***a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.***

The Fleets-As-Areas (FAA) assessment has many more parameters to be estimated than the single area assessment and the assessment team spent considerable time explaining the choice of the current FAA model structure and looking carefully at the model inputs to determine the contributions they made to the model stability and model outputs.

Trawl-based indices. The trawl survey provides a long time series from 1984 to 2025, covering the upper continental shelf to a depth of 500 m. Prior to 2007, it was not used in the assessment because it was considered a poor indicator of adult sablefish abundance. Adult sablefish generally occur farther down the slope and may also be able to outswim the trawl gear.

From 2007 to 2024, the trawl index was included in the assessment because it was considered to provide useful information on recruitment. However, over the past decade in the Gulf of Alaska (GOA), the index has shown substantial fluctuations. It was very low in 2013, very high in 2021, and then declined sharply in 2023. These changes, particularly the

2023 decline, were not consistent with trends in the Longline Survey indices, making the trawl index an uncertain source of information on sablefish abundance.

The assessment team also noted that, in recent years, younger sablefish, aged 2–4 years may have moved and resided farther down the slope to depths greater than 500 m, reducing their availability to the trawl survey. In addition, the Aleutian Islands (AI) and Bering Sea (BS) trawl indices have high sampling error.

The model runs with and without the trawl indices produced similar model outputs, for example, spawning biomass. Across AI, BS, and GOA, the trawl indices appeared to add data volume but did not provide additional information beyond that already contained in the Longline Survey indices. Consequently, the trawl survey data were excluded from the assessment.

Overall, the model results indicate that removing the trawl survey data produces a more parsimonious assessment without materially changing key outputs.

Recommendation 4.1.1. I concur with the decision to discontinue use of the trawl survey data in the 2026 assessment.

NOAA Longline Survey. This survey provides a continuous 35-year time series sampling from habitat occupied by sablefish. It tracks both adult and juvenile abundance and shows broadly consistent signals across the Aleutian Islands (AI), Bering Sea (BS), and Gulf of Alaska (GOA).

From 1979 to 1994, the survey was conducted as a Japanese–United States cooperative programme that provided information on relative population numbers, lengths, and ages. From 1990 to 2025, NOAA conducted its own survey, with the overlapping years providing an important calibration period between the two data sources.

The Longline Survey is a fixed-station survey in which, as far as possible, the same sites are fished each year. The number of sites varies by region. Each site typically includes eight skates in the GOA, with fewer skates sometimes in the BS and AI because of habitat and logistical constraints. Each skate has 45 hooks and samples depths of 200, 400, 600, 800, and 1,000 m. In general, otoliths are collected from one fish per skate.

The purpose of the survey is to track changes in population numbers across the fishery, not to determine spatial estimates of biomass. In my view, the survey achieves this objective successfully, as shown by the consistency of Relative Population Number (RPN) values among regions and their agreement with the age composition data.

Adjustments to the Longline Survey data. Adjustments were made in three circumstances:

- Damaged hooks. If more than five hooks on a skate were damaged and therefore not fishable, catch numbers were adjusted on a pro rata basis.

- Sperm whale interactions. If a sperm whale was observed near the vessel and appeared to be interacting with the gear, the skipper and observer determined whether the skate had been affected. Where an effect was identified, a correction factor of 1.18 was applied to the catch for that skate. This adjustment is more commonly applied in the Gulf of Alaska (GOA) than in other regions, but its overall effect is small, typically around 1–5%.
- Killer whale interactions: Killer whale interactions can cause unpredictable and sometimes substantial fish loss from individual skates. If one or more killer whales were observed near the vessel, the skipper and scientist jointly assessed whether the skate had been compromised. If so, the skate was removed from the dataset.

Recommendation 4.1.2. There are guidelines, for the skipper and observer to follow on whether a skate is removed from the dataset. It would be valuable to develop a more formal and agreed protocol to decide if a skate was removed or retained after a killer whale interaction. Given that the adjustment is only 1-5%, this is not a great concern.

From the start of the NOAA survey until 2023, the AI and BS were surveyed biennially, while the GOA was surveyed annually. During this period, the GOA index was used to extrapolate the AI and BS indices in years when those regions were not surveyed. The Relative Population Number (RPN) for each region, AI, BS, and GOA, is calculated as the summed number of sablefish observed in that region during the survey.

In 2024, obtaining a suitable vessel for the Longline Survey became difficult. One vessel was available to conduct the GOA survey for 50 days every second year. This led to a revised design in which the BS and AI are surveyed in even years, and the GOA is surveyed in odd years, with three GOA survey sites being removed. As a result, the GOA survey index for the missing year was filled using the value from the previous year, and the AI and BS survey indices continued to be extrapolated as before. This was an undesirable but necessary interim approach under the revised survey constraints.

Implications of biennial GOA survey coverage. A major concern is the loss of GOA survey coverage in every second year. Replacing the missing GOA index with the previous year's value is problematic and needs to be addressed. In the single-area assessment model, there is no satisfactory solution. By contrast, the Fleets-As-Areas (FAA) assessment allows the even-year data for AI and BS, and the odd-year data for GOA, to be treated as missing observations.

Recommendation 4.1.3. Moving from the single area model to a FAA assessment is highly recommended as it provides a defensible way to accommodate the revised survey design in GOA.

Longline Survey as a numbers-based index. The LLS index for each region AI, BS, and GOA is expressed as Relative Population Numbers (RPN), a numbers-based index. In years of strong recruitment, the RPN can be dominated by small fish. This seems peculiar, making me wonder whether it might be preferable to convert fish numbers to biomass using a weight-at-age relationship. The assessment team had already evaluated this option and concluded that converting numbers to weight would just add uncertainty to the assessment outputs.

Recommendation 4.1.4. This may be worth further consideration, but as a low priority, because the age composition data already provide information on the proportions of younger and older fish and the conversion of lengths to numbers just increases uncertainty.

CPUE indices. These indices from logbook data were not available to NOAA after 2022. In any case, these data have become increasingly unsuitable for assessment purposes because the stock has changed rapidly since 2015 following several strong recruitment events. The fishery has also undergone major gear changes in recent years, shifting from hook-and-line gear towards pot fishing, particularly through the use of collapsible “slinky” pots that can be carried on smaller vessels.

The CPUE index does not match the Longline Survey index because the fishery tends to target larger fish. This effect has been particularly evident in recent years, following several strong recruitment events.

The Japanese CPUE data from the early period of the fishery were also removed because they could not be standardised and because the provenance of the data was uncertain.

Recommendation 4.1.5. Discontinuing the use of both sources of CPUE data is recommended as it improves model stability and produced better fits in the retrospective analyses.

Age composition data. These data provide important information on recruitment patterns and the age structure of the stock. In the Gulf of Alaska (GOA), otoliths have been collected annually since 1990. Across the Western, Central, and Eastern Gulf areas, the annual age composition sample generally comprises approximately 600–1,200 fish. In the Aleutian Islands (AI) and Bering Sea (BS), age samples are collected every second year. Sampling began in the AI in 1996, with variable annual sample sizes of approximately 140–300 fish, and in the BS in 1997, where sample sizes are smaller and more variable, typically around 50–100 fish.

These data are particularly valuable for identifying strong recruitment events. They indicate strong recruitment in the AI in 2014, 2016, and 2017; in the BS in 2017; and in the GOA in 2014 and 2016.

The assessment model showed considerable tension between the NOAA Longline Survey (LLS) data and the age composition data. When the age composition data were fitted with a multinomial likelihood, the fit to the LLS data deteriorated. This required an adjustment to the effective sample size for the age composition data.

The assessment team used Francis TA1.8 iterative reweighting, which generally reduces the sample size, giving more weight to the LLS index which generally has lower variance, and less weight to the age composition in the likelihood. This adjustment improved the fit to the survey indices while retaining satisfactory fits to the age composition data. The Francis reweighting is a common and accepted approach, and I agree with its continued use in the sablefish assessment.

The assessment team also explored a Dirichlet–multinomial likelihood to determine whether Francis reweighting could be avoided and whether the tension between the two datasets could be reduced. The resulting fits were unsatisfactory, and this approach was not pursued.

The plus-group. The model fits to the age composition data are generally acceptable, particularly given the longevity of sablefish and the relatively sparse observations in individual age classes. The most conspicuous issue is the large plus group (ages 31 and older). This is especially noticeable in the AI from 2010 to 2016, when large aggregations of older fish were observed. The model estimates for the plus group are substantially below the observed values, and this component of the data cannot be fitted well.

There was discussion at the Review Meeting to remove the plus group entirely. The Assessment Team had already tested this, and key model outputs, including spawning biomass, changed little.

Recommendation 4.1.6. The plus-group should remain in the age composition model input. Although it is visually undesirable, this is not a reason to delete it.

In the FAA model, the issues of tension between the age composition and LLS data, reliance on Francis's reweighting, and poor fit to the plus group, are more serious than in the single-area model because the data are distributed across three regions. Despite these difficulties, the overriding consideration is how best to accommodate the missing years of LLS data. In this respect, the FAA model has a clear advantage over the single-area model because it can treat the missing regional observations as missing data and adjust the model fitting accordingly.

Selectivity. Logistic selectivity is used in all cases for AI, BS, and GOA. Looking at the data, common sense would indicate that a dome shaped selectivity could be more appropriate in some cases. The dome-shaped selectivity was investigated in detail, and it was found that the parameters were difficult to estimate, and the model was unstable compared to the logistic selectivity. The added complexity did not give greatly different estimates of the model outputs, and a simple, more parsimonious model structed was selected.

The assessment team investigated changes to selectivity time blocks. More or different time blocks resulted in model instability and these formulations were rejected.

Recommendation 4.1.7. In the short term, logistic selectivity in the current time blocks is recommended for the assessment, as it is the most parsimonious parameter configuration and has more stable estimation and similar model outputs.

Recommendation 4.1.8. There should be further investigation into different time blocks for the FAA assessment, to see if there are other formulations that result in stable parameter estimation. The use of sex-specific spline selectivity could be investigated. Possibly 4 nodes and the same time blocks as the current logistic selectivity. The extra number of parameters may well be manageable in the FAA model.

Recommendation 4.1.9. The change in the survey design makes the use of the single area model untenable and I recommendation that the FAA model be used for the current sablefish assessment.

4.2 Terms of Reference 2

Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.

Advise as to whether the model is consistent with and represents the best scientific information available (BSIA) for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

Conceptualization of the FAA model. The major drawback of the FAA model is the interpretation of the assessment process. In this assessment, it is assumed there is a single homogeneous population and the differences in the Longline Survey indices and age composition data across AI, BS, and GOA are interpreted as due to changes in the availability of sablefish to the fleet rather than biological differences across the fishery, for example recruitment. The model fitting to the signals coming from the indices and age composition come through estimates of q and selectivity.

Recommendation 4.2.1. When reporting the outputs of the assessment to Managers and Stakeholders care needs to be taken to keep it simple and not allude to the conceptual difficulties of the FAA assessment model.

Parameter Estimation. The single area model (26.1_1Reg) has 253 estimated parameters, and the fleets-as-areas model (26.2_FAA) has 271 estimated parameters. The extra parameters in the FAA model are 2 catchability parameters and 16 selectivity parameters.

Jitter Analysis. The Jitter analysis showed, to my great surprise, that the FAA model converged 100% of the time.

Recommendation 4.2.2. The Jitter analysis should be revisited with larger perturbation of the starting values.

Natural Mortality. The model input value of natural mortality (*M*) is highly influential in the model outputs, like spawning biomass. The value of *M*=0.085 was selected based on tagging models. The SAFE report mentions this is associated with maximum age of about 65 years, considered appropriate because few aged fish were greater than 70 years.

In the FAA assessment model, natural mortality is estimated with a prior of *M*=0.085 for males and females, with a small cv. This allows for the error in *M* to be incorporated in the assessment, while constraining it to be close to the best available estimate.

The higher value of natural mortality used in previous assessments, will result in the assessment model determining there are fewer older fish, resulting in lower spawning biomass, lower B/B_{40}, lower B/MSY, and ABC and OFL will decrease.

Recommendation 4.2.3. The choice of 0.085 as the prior for *M* with a small cv is appropriate. In this assessment, *M* is age invariant. I believe that varying *M* with age is un-necessary and unlikely to influence the model outputs. The comment in the SAFE report on the how 0.085 relates to maximum age of 65 was confusing. A brief note on how 65 was derived would be useful.

Fits to the NOAA Longline Survey data. The FAA model has serious tension between the fitting of the NOAA Longline Survey data and the age composition data. This is common and not unexpected. After the Francis iterative re-weighting, the fit to the survey indices improved. The survey data does not provide coverage of the whole sablefish population especially in AI. This inadequacy is common to the single area, FAA, and spatial modelling approaches and will not change.

Age composition The aim is to age approximately 1200 otoliths per year with 300 from AI and BS in alternate years and in GOA 300 from each of East, Central and West. This sample size is possibly large enough to get acceptable age compositions for the single area model.

In the FAA assessment, the sample size of 300, spread over two sexes and ages classes 2 to 70 is barely adequate. There is potential for improving the age composition data without greatly increasing the management cost of the fishery.

The “AFSC Aging Methods Database” lists the number of sablefish otoliths collected and the number aged. An extract from the table is shown below.

	AI		BS		GOA	
Year	Aged	Collected	Aged	Collected	Aged	Collected
2025	0	0	0	0	0	3051
2023	0	0	300	562	886	3643
2022	298	397	0	0	895	3378
2021	0	0	300	715	889	3951
2020	296	524	0	0	890	2223
2019	0	0	299	954	894	4300
2018	297	441	0	4	891	1778
2017	0	0	300	411	890	2930
2016	303	406	0	455	892	1832

This data indicates that there are many otoliths collected and not aged in all regions of the fishery. The collection of the otoliths is, I expect, a much higher cost than the aging. Sablefish are extremely difficult to age because of the small differences in seasonal growth due to low temperature variation where the sablefish live. Thus, deciding which annuli are annual growth rings is difficult. It is not a simple matter to find a new person to age sablefish as considerable training and evaluation is necessary.

Recommendation 4.2.4. An investigation should be conducted with the FAA model, of the effect on model stability and model outputs with changes in sample size (like a random sample of half the ages).

Recommendation 4.2.5. The GOA survey changing to every second year may allow more resources for otolith reading. I recommend another otolith reader be employed and trained. Finance might come from the reduced NOAA Longline Survey coverage in GOA. Industry could be approached and asked if they would like to contribute to reading more sablefish otoliths.

Recommendation 4.2.6. As the sablefish otoliths are extremely difficult to age, a new reader could be allocated to reading some of the easily read species, freeing up an experienced sablefish reader to take on more sablefish otolith reading.

Closed loop simulation. The single area model must extrapolate the RPN when the GOA index it not available, resulting in a systemic bias in biomass and recruitment. The closed loop simulation indicated that with the 16 extra parameters, the FAA model was able to better account for the signals in the data, had lower spawning biomass bias, and had greater model stability.

Projections and Management advice. At the present time, the FAA model provides the best information on the state of the Alaska-wide sablefish stock. The FAA model generates reliable projections which are used for estimating the required Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

Recommendation 4.2.7. The FAA model has a huge advantage over the one area model as it can reliably accommodate the problems with the alternate year missing Longline Survey index data in GOA. The FAA model should be adopted for the current assessment.

4.3 Terms of Reference 3

Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Conceptual Understanding. The spatial assessment model has clear conceptual advantages over the FAA model. The Longline Survey data, age composition, and tagging data is collected on a spatial scale across AI, BS, and GOA and the regional recruitment and movement can be modelled on a spatial scale. This avoids the use of fleet catchability and selectivity to fit the data, which is difficult to conceptualize.

The Longline Survey indices have recently been generally higher in GOA, especially centre and east, indicating that GOA is likely to be the dominant area of the fishery. The larger number of age composition samples in GOA will bias the model fitting to favour GOA data in the FAA model. These problems can be avoided in a 3-area, or better still a 5-area spatial model.

The high recruitment years in AI, BS, and GOA do not occur in the same years. This cannot be modelled in the FAA model with the assumption of a single sablefish population across the whole area of the fishery. With the spatial model, the recruitment can be estimated in each area. This is a far more common-sense approach, allowing for different recruitment in each area.

The current 5-area model structure is a sex-specific, age-structured model with

recruitment estimated in each area with
movement across areas estimated in 3 age blocks over 3 spatially invariant time blocks (pre-1990, 1990-2015, and 2016 to present),
sex-specific but time-invariant logistic selectivity
catchability common over areas
recruitment estimates in each area.

The model has 1208 parameters and takes 25 minutes to run.

In the 5-area model, the fit to the indices is generally good. The age composition is acceptable but not as good in AI and BS as GOA, especially the plus group. The smaller number of aged fish in these two areas has exacerbated this. The fits to the tagging data in each of the 5 areas is good, especially in the early years. Changes in the fishing gear to pots could be contributing to poorer fit in the recent years.

It is clear, especially recently, that the timing of the high recruitment events is different across the fishery, in AI, BS, and GOA. The FAA model unrealistically treats the fishery as one population with a fishery wide recruitment.

For the 5-area model, the retrospective analysis showed positive retrospective patterns indicating model instability. In many cases Mohn's Rho was close to 0.2 and in BS Mohn's Rho >0.2.

Tagging data and movement. The critical additional data required for the 5-area spatial model is the tagging data. This gives us information on the age-specific movement of juveniles, sub-adults and adults, across the regions. The model configuration was to calculate the movement between adjacent regions in 3 age blocks.

Selectivity. Logistic selectivity was used throughout for the 5-area spatial model. All the problems with selectivity in the FAA model exist, but on top of that we have the problem of more selectivity parameters causing instability in the parameter estimation. Loosening up the priors or constraining them tightly were not solutions. Alternative selectivity formulations are required before the spatial model becomes a viable alternative to the FAA model.

Recommendation 4.3.1. Alternative selectivity formulations (like a 4-node spline) need to be explored in the hope of moving to a spatial model in the future.

Data sparsity. The NOAA Longline Survey has reduced coverage in GOA through a reduced number of stations after 2024 and the change to a biennial Longline Survey in GOA. This has increased uncertainty in the model outputs.

Under the current regime, during the NOAA Longline Survey, otoliths will be collected in GOA every second year. It should be clarified if the plan to collect about 3000 otoliths every second year or to maintain 3000 per year.

Recommendation 4.3.2. The NOAA Longline Survey is the only available measure of population trends. Further reductions in the survey coverage would seriously impact the model outputs and must be resisted.

Recommendation 4.3.3. The otolith collection across the fishery should be at least that before the changes to the survey regime GOA.

Recommendation 4.3.4. With a 5-area spatial model, the age composition data will be more sparse in GOA and I recommend that more otoliths need to read to improve the information on age structure and especially recruitment, to inform the assessment model.

Recommendation 4.3.5. The 5-area model makes sense conceptually as it captures the movement and age composition information in the fishery more sensibly. Development of this spatial model should continue.

The 3-area model. This model configuration with areas AI, BS, and GOA had a similar structure to the 5-area model with 723 parameters to be estimated. In general, the fits to the age composition data and the survey indices were like the 5-area model. The retrospective analysis showed concerning retrospective patterns like the 5-area model.

Understanding the model structure. An important positive for the 5-area assessment is that stakeholders, Managers, and Scientists can more easily understand the concepts of biomass in each region, movement, different regional recruitment, and regional increases or decreases in biomass. The allocation of quota, based on the Longline Survey indices in each region is easier to understand in a spatial model because it can be understood that is based on the biomass in each area.

The 5-area model has advantages over the 3-area model as it can potentially pick up regional depletion, matched more clearly with the apportioning of quotas in AI, BS, EG, CG, and WG.

Reporting of projections. The 5-area, and 3-area model show high instability in estimation of the recruitment, movement, and selectivity parameters due to overparameterization and low sample sizes. Although the model configuration needs more flexible selectivity to fit the age composition data, this is not possible due to the limited data. The projections are unreliable as are the estimates of the reference values required to determine the status of the fishery.

Movement Simplification. A possible way to reduce the number of estimated parameters in the 3-area or 5-area area assessment model is to determine the movement across the areas outside the model and bring them in as fixed parameters. This was tried by the assessment team, but the problem is that movement needs to be age specific making this an untenable solution.

Longline Survey Tagged fish. About 5% of the Longline Survey sablefish are tagged and released, about 3000-3500 per year. This amounts to 413,000 sablefish tagged and 38,000 tags recovered. In general, there are no otoliths associated with these tag recoveries as the tags are recovered by fishers and commercial landings. This data is valuable for movement patterns and is vital information for a 5-area or 3-area assessment model

Juvenile tagging. AFSC conducts a tagging program in GOA where 42,625 juvenile sablefish were tagged, with conventional tags, and the number of returns with tags and otoliths is 2630. The otolith recoveries generally come from

- an observer at a plant
- an observer on a vessel,
- a tagged fish being caught during a survey.

The number of tag returns from this program without otoliths is not clear.

This program supplies otoliths of known age, enabling the development of an aging error matrix which is directly incorporated in the assessment model to account for aging error in the model outputs. In addition, this data contributes information on age specific movement.

Recommendation 4.3.6. It is interesting to have a large quantity of information on reader error due to having so many fish of known age. I would like to see if a reduced aging error would improve model stability and how model outputs change.

Longline Survey tag returns. About 500 to 1000 tags are recovered each year from fishers and processors. A recovered tag entitles the individual to a beanie and an entry in the lottery to win $1000. I believe there is potential to gain more aging information from both tagging programs at little extra cost.

Recommendation 4.3.7. The number of otoliths recovered from tagged fish should be encouraged. Tag return incentives could be changed to give the fisher or the processor greater incentive to return the tag with otoliths. Increase the number of shares in the lottery to 5 or 10 shares for a tag with fish frame or otoliths. The prize money could also be increased from $1000 to $2000.

Otolith Staining. Injecting calcein into tagged fish is not a health concern (unlike oxytetracycline) and would greatly contribute to the fish aging process, possibly reducing aging error.

Recommendation 4.3.8. A sample of fish tagged on the Longline Survey be injected with calcein to assist in identification of annual annuli.

Reporting of Reference Points. The Alaska-wide sablefish requires Tier3 reference points. The projections are used to report F_{ABC} defined as $F_{40\%}$ and F_{OFL} defined as $F_{35\%}$. For a 5-area spatial model, where biomass and spawning biomass is estimated in each of the 5 areas, it would make sense to report the reference points in each area. Currently, there are no guidelines on reporting of area specific reference points. With a spatial assessment, it is not clear how F_{ABC} and F_{OFL} would be determined.

Recommendation 4.3.9. The mismatch between the Management requirements and the area specific reference points should be investigated and resolved.

3-area spatial assessment. This assessment model has 723 estimated parameters, compared with 1208 for the 5-area model. In the 3-area spatial model the fit to the survey indices was good and there was an adequate fit to the age composition, except in the middle years. There are moderate retrospective patterns in both the 3-area and the 5-area models. It is slightly worse in BS for the 3-area model and slightly worse in GOA in the 5-area model. The problems involved in determining suitable selectivity formulations, parameter estimation, and model stability, are similar for the 3-area and 5-area assessment.

Recommendation 4.3.10. The Alaska-wide assessment should move towards a spatial model when it is sufficiently stable to provide reliable projections for calculating the reference points. Given the considerable variation in catch across east, middle and west GOA (highest in the east), I recommend that future work on the spatial model should be with a 5-area model, rather than a 3-area model.

Appendix 1 Documents provided for the review.

Core documents

1. Appendix 3A_SPoRC_AssessmetDescription.pdf
2. Appendix 3B_1_Reg_Model_Build_Final.pdf
3. Appendix 3C_FAA_Model_DevelopmentModel_Build_Final.pdf
4. Appendix 3D_Spatial Model _Final.pdf
5. Appendix 3E_Closed_Loop_Sim_Final.pdf
6. Sablefish_SAFE_CIE_2026_Final.pdf

Additional materials supplied for the meeting

1. The 2024 Alaskan sablefish Stock Assessment and Fishery Evaluation (SAFE) report (Goethel and Cheng, 2024) and presentation;
2. The 2024 December SSC (p. 14-16) and November Joint Plan Team (p. 2-3) reports;
3. The 2025 AFSC Longline Survey Cruise Report (Siwicke et al., 2026) and presentation;
4. The 2025 October SSC (pp. 21-23) and September Joint Plan Team (p. 5-6) reports;
5. The SPoRC assessment platform publication (Cheng et al., 2026) and GitHub documentation;
6. Research papers on integrating density-dependent growth (Cheng et al., 2025b) and a pot gear fleet (Cheng et al., 2024a) into the sablefish assessment, simulation studies on selectivity (Cheng et al., 2024b) and sex-specific parametrization (Cheng et al., 2025c) for the sablefish assessment, and considerations for fitting sex-specific composition data (Cheng et al., 2025d) in stock assessments;
7. CIE reviewer reports from the 2016 Alaskan sablefish CIE review.

Model inputs

1. https://github.com/dgoethel-noaa/2026_Sablefish_CIE/tree/main/Model Runs/1Reg_Mods
2. https://github.com/dgoethel-noaa/2026_Sablefish_CIE/tree/main/Model Runs/FAA_Mods

Appendix 2: Performance Work Statement (PWS)

National Oceanic and Atmospheric Administration (NOAA)
NOAA Fisheries
Center for Independent Experts (CIE) Program
External Independent Peer Review

Alaska-wide Sablefish Stock Assessment
June 16 - 18, 2026

Background

NOAA Fisheries is mandated by the Magnuson-Stevens Fishery Conservation and Management Act, Endangered Species Act, and Marine Mammal Protection Act to conserve, protect, and manage our nation's marine living resources based upon the best scientific information available (BSIA). NOAA Fisheries science products, including scientific advice, are often controversial and may require timely scientific peer reviews that are strictly independent of all outside influences. A formal external process for independent expert reviews of the agency's scientific products and programs ensures their credibility. Therefore, external scientific peer reviews have been and continue to be essential to strengthening scientific quality assurance for fishery conservation and management actions.

Scientific peer review is defined as the organized review process where one or more qualified experts review scientific information to ensure quality and credibility. These expert(s) must conduct their peer review impartially, objectively, and without conflicts of interest. Each reviewer must also be independent from the development of the science, without influence from any position that the agency or constituent groups may have. Furthermore, the Office of Management and Budget (OMB), authorized by the Information Quality Act, requires all federal agencies to conduct peer reviews of highly influential and controversial science before dissemination, and that peer reviewers must be deemed qualified based on the OMB Peer Review Bulletin standards[1].

Sablefish (*Anoplopoma fimbria*) are a long-lived (90+ years), highly mobile, sexually dimorphic demersal species that live at depths greater than 200m (as adults). In Alaska Federal waters, sablefish are assessed and managed as a single population with the total quota apportioned across 6 management regions (Aleutian Islands, Bering Sea, Western Gulf of Alaska, Central

[1] https://www.whitehouse.gov/wp-content/uploads/legacy_drupal_files/omb/memoranda/2005/m05-03.pdf

Gulf of Alaska, Western Yakutat, and Southeast Outside) and among two fishing sectors (i.e., fixed gear and trawl gear). Since rationalization (i.e., implementation of individual fishing quotas, IFQs) occurred in the 1990s, sablefish have been one of the most valuable finfish resources in Alaska ($124 million in ex-vessel value in 2022). Catch advice for sablefish is based on a statistical catch-at-age integrated stock assessment model fit to fishery-independent survey data and fishery-dependent catch-age data. The sablefish assessment is reviewed annually by the North Pacific Fishery Management Council's (NPFMC) Science and Statistical Committee (SSC) as part of the quota specification process and last underwent a CIE review in 2016.

Scope

Since the last CIE review (i.e., in 2016), major changes have occurred in the sablefish resource, fishery, longline survey, and assessment model. Specifically, a series of extreme, above average recruitment events has led to rapid rebuilding of the resource and increasing quotas. Resultant landings of predominantly small, lower value fish have led to market saturation, strong reductions in landed value, and exploration of management options that loosen full retention requirements. Concomitantly, the primary directed fixed gear fleet has undergone rapid gear changes, where a predominantly hook-and-line fishery has transitioned to a longline pot fishery since 2017 (i.e., >85% of sablefish catch in the fixed gear fishery is now taken by pot gear), which has helped to reduce whale depredation.

Additionally, the fishery-independent, cooperative research-based NOAA longline survey, which targets sablefish and serves as the primary index of abundance for the assessment, underwent major survey design changes starting in 2025. Because the survey has traditionally used a cost-recovery approach, whereby survey catch is sold by the fishing company that owns the vessel performing the survey, the decline in sablefish markets prevented the survey from being undertaken in 2024 for the first time in its 30+ year history. Historically, the fixed station survey has sampled the entire Gulf of Alaska (GOA) annually, as well as either the Bering Sea (BS, odd years) or Aleutian Islands (AI, even years). Starting in 2025 a new survey design was enacted to reduce overhead and ensure a fishing company would accept the survey contract, wherein the Gulf of Alaska is now surveyed in odd years (starting in 2025), and the BS and AI are surveyed in even years (starting in 2026). Although the new survey design ensures the ability to continue the longline survey, it results in only ~50% of total stock abundance being surveyed in a given year. Moreover, it raises important challenges for the panmictic assessment in terms of how to integrate survey data that only partially samples the population being assessed.

There have also been major changes to the assessment model since the last CIE review. Most importantly, a new R-TMB based package, the SPoRC (Stochastic Population Over Regional Components) assessment model, has been adopted, which replaces the previous Automatic Differentiation Model Builder (ADMB) based assessment model. Additionally, a number of research-oriented assessment models have been developed exploring alternative spatial structures (e.g., spatially-explicit 3 and 5 region models that estimate movement among regions, as well as a spatially-implicit single region Fleets-As-Areas (FAA), all of which demonstrate promise for dealing with spatial complexities and data gaps that are not well resolved in the single region panmictic assessment. In 2026, the assessment team transitioned from the single region panmictic assessment to the FAA model, because the FAA model could better handle the new longline survey design (i.e., fit the data by region as separate fleets) and implicitly account for ontogenetic movement dynamics (i.e., through selectivity and availability assumptions). Thus, for the purpose of the CIE review, feedback on the FAA model is being requested, particularly with regard to appropriate model structure and parametrization. To help make determinations on the BSIA to assess the Alaska sablefish population, reviewers will be provided the results of closed loop simulation analyses and assessment model runs applied to the sablefish data (including model outputs and diagnostics).

The overarching goals of this review process are to:

1. Ensure the FAA stock assessment supports the development of robust advice for use by the North Pacific Fishery Management Council based on the BSIA;
2. Provide an independent external review of this stock assessment to confirm that it meets the mandates of the Magnuson-Stevens Fisheries Conservation and Management Act (MSA) and other legal requirements;
3. Understand and identify appropriate FAA model structure and fits to key data sources (indices, landings, and age compositions);
4. Identify research needed to improve the assessment and advice for fishery managers.

The specified format and contents of the individual peer review report are found in **Annex 1**. The Terms of Reference (ToRs) of the peer review are included in **Annex 2**. The tentative agenda of this in-person panel review meeting is included in **Annex 3**.

Requirements

NOAA Fisheries requires **3 reviewers** to conduct an impartial and independent peer review in accordance with this Performance Work Statement (PWS), OMB Guidelines, and the ToRs below. Modifications to this Performance Work Statement (PWS) and ToRs cannot be made during the peer review, and the Contracting Officer's Representative (COR) and the CIE contractor shall approve any modifications prior to the peer review. All ToRs must be addressed in each reviewer's report.

The CIE reviewers should have expertise in statistical sex- and age-structured integrated stock assessment models. In particular, experience developing spatially-explicit and spatially-implicit (i.e., fleets-as-areas) assessment models is imperative. Moreover, understanding of stock assessment good practices are required. Working knowledge of the RTMB programming language, a basic understanding of closed loop simulation, and a background in survey design and/or the use of fishery-independent survey data in stock assessments would all be beneficial.

The chair for this meeting, who is in addition to the CIE reviewer, will not be provided by the CIE. Although the chair will be participating in this review, the chair's participation (e.g., labor and travel) is not covered by this contract.

Tasks for Reviewers

The CIE reviewer shall complete the following tasks in accordance with the PWS and Schedule of Milestones and Deliverables herein.

Pre-review Background Documents: A minimum of two weeks before the peer review, the NOAA Fisheries Project Contact will send (by electronic mail or make available at an FTP site/GitHub Repository) to the CIE reviewers the necessary background information and reports for the peer review. In the case where the documents need to be mailed, the NOAA Fisheries Project Contact will consult with the CIE on where to send documents. The CIE reviewers are responsible only for the pre-review documents that are delivered to the reviewer in accordance with the PWS scheduled deadlines specified herein.

The document list is separated into 'summary/background' and 'for review' documents. The CIE reviewers are *encouraged to peruse* the 'summary/background' documents for information on data inputs, previous assessment structures and associated reviews, and ongoing research. However, the CIE reviewers *must review* all of the 'for review' documents in preparation for the peer review, as these contain the results of the primary analyses and model runs that will be discussed during the review panel.

Available documents (available from the public sablefish CIE GitHub repository, as they become available) include:

For Review (available here)

1. A summary document describing the fleets-as-areas assessment model, including the full model description and results, with comparison to outputs from the single region panmictic assessment as well as summary results for all sensitivity model runs;
2. A document summarizing the results of the spatially-explicit closed loop simulation analysis;
3. The 2025 Alaskan sablefish stock assessment proposed model changes report (Goethel and Cheng, 2025) produced for the September 2025 Joint Plan Team meeting, presentation, and GitHub repository, which describes recent changes to the current single region panmictic assessment including transition to RTMB and the SPoRC assessment package;
4. The spatial sablefish model publication and model updates report (Cheng et al., 2025a) for the September 2025 Joint Plan Team meeting;

5. Additional supporting documents and presentations as they become available.

Summary/Background (Available here)

1. The 2024 Alaskan sablefish Stock Assessment and Fishery Evaluation (SAFE) report (Goethel and Cheng, 2024) and presentation;
2. The 2024 December SSC (p. 14-16) and November Joint Plan Team (p. 2-3) reports;
3. The 2025 AFSC Longline Survey Cruise Report (Siwicke et al., 2026) and presentation;
4. The 2025 October SSC (pp. 21-23) and September Joint Plan Team (p. 5-6) reports;
5. The SPoRC assessment platform publication (Cheng et al., 2026) and GitHub documentation;
6. Research papers on integrating density-dependent growth (Cheng et al., 2025b) and a pot gear fleet (Cheng et al., 2024a) into the sablefish assessment, simulation studies on selectivity (Cheng et al., 2024b) and sex-specific parametrization (Cheng et al., 2025c) for the sablefish assessment, and considerations for fitting sex-specific composition data (Cheng et al., 2025d) in stock assessments;
7. CIE reviewer reports from the 2016 Alaskan sablefish CIE review.

Please note that outputs of all analyses used as part of the CIE review, including input data, simulation outputs, final proposed assessment models, and all sensitivity assessment models, will be made available on the sablefish CIE GitHub repository

Panel Review Meeting: The CIE reviewers shall conduct the independent peer review in accordance with the PWS and ToRs and shall not serve in any other role unless specified herein. The CIE reviewers shall actively participate in a professional and respectful manner as members of the meeting review panel, and their peer review tasks shall be focused on the ToRs as specified herein. The NOAA Fisheries Project Contact is responsible for any meeting arrangements (e.g., conference room reservations, video conferencing logistics, etc.).

Contract Deliverables - Independent CIE Peer Review Report: Each CIE reviewer shall complete an independent peer review report in accordance with this PWS. Reviewers are not required to reach a consensus, and a summary report *will not* be produced. Each CIE reviewer shall complete their independent peer review according to the required format and content as described in **Annex 1**. Each CIE reviewer shall complete the independent peer review addressing each ToR as described in **Annex 2**. The tentative agenda of the panel review meeting is in **Annex 3**. Each CIE reviewer will deliver their reports according to the specified milestone dates.

Foreign National Security Clearance

When reviewers participate during a panel review meeting at a government facility, the NOAA Fisheries Project Contact is responsible for obtaining the Foreign National Security Clearance approval for reviewers who are non-US citizens. For this reason, the reviewers shall provide requested information (e.g., first and last name, contact information, gender, birth date, passport number, country of passport, travel dates, country of citizenship,

country of current residence, and home country) to the Project Contact for the purpose of their security clearance, and this information shall be submitted at least two weeks in advance.
For additional information, please see the following link: https://www.commerce.gov/osy/programs/foreign-access-management. All CIE reviewers must have a REAL ID-compliant form of identification to access Federally owned or leased facilities. The contractor is required to use all appropriate methods to safeguard Personally Identifiable Information (PII).

Place of Performance
The place of performance shall be Juneau, AK.

Period of Performance
The CIE reviewers' duties shall not exceed **14** days to complete all required tasks.

Schedule of Milestones and Deliverables
The contractor shall complete the tasks and deliverables in accordance with the following schedule:

Schedule	Milestones and Deliverables
Within two weeks of award	Contractor secures participation of the CIE reviewers.
At least two weeks prior to the panel review meeting	Pre-review documents are provided to the reviewers.
June 16 - 18, 2026	The reviewers participate and conduct an independent peer review during the panel review meeting.
Approximately 2 weeks later	Contractor receives draft reports.
Within 3 weeks of receiving draft reports	Contractor submits final reports to the US Government.

Applicable Performance Standards
The acceptance of the contract deliverables shall be based on three performance standards: (1) The reports shall be completed in accordance with the required formatting and content in **Annex 1**; (2) The reports shall address each ToR as specified **Annex 2**; and (3) The reports shall be delivered as specified in the schedule of milestones and deliverables.
Travel

All travel expenses shall be reimbursable in accordance with Federal Travel Regulations (Travel resources | GSA), and all contractor travel must be approved by the COR prior to the actual travel. Any travel conducted prior to the receipt of proper written authorization from the COR will be done at the Contractor's own risk and expense. International travel is authorized for this contract. Travel is not to exceed $13,000.00.

Confidentiality and Data Privacy

This contract may require that services contractors have access to Privacy Information. Services contractors are responsible for maintaining the confidentiality of all subjects and materials and may be required to sign and adhere to a Non-disclosure Agreement (NDA).

NOAA Fisheries Project Contact:

Chris Lunsford
MESA Program Manager
NOAA Fisheries, Alaska Fisheries Science Center
17109 Pt. Lena Loop Road
Juneau, AK 99801
Chris.Lunsford@noaa.gov

Annex 1. Peer Review Report Requirements

1. The CIE independent report shall be prefaced with an Executive Summary providing a concise summary of the findings and recommendations as specified in the ToR and provide advice on whether the science reviewed is consistent with and represents the best scientific information available (BSIA).
2. The main body of the reviewer report shall consist of the following sections: Background; Description of the Individual Reviewer's Role in the Review Activities; Summary of Findings for each ToR in which the weaknesses and strengths are described; and Conclusions and Recommendations in accordance with the ToRs.
    a. Each reviewer should describe in their own words the review activities completed during the panel review meeting, including providing a brief summary of findings as well as the science, conclusions, and recommendations.
    b. Each reviewer should discuss their independent views on each ToR even if these were consistent with those of other panelists, and especially where there were divergent views.
    c. Each reviewer should elaborate on any points that they feel might require further clarification.
    d. Each independent CIE report shall be a stand-alone document through which others can understand the weaknesses and strengths of the science reviewed and shall be an independent peer review of each ToRs.
3. The reviewer report shall include the following appendices:

Appendix 1: Bibliography of materials provided for review.
Appendix 2: A copy of the CIE Performance Work Statement.
Appendix 3: Panel Membership or other pertinent information from the panel review meeting.

Annex 2. Terms of Reference (ToRs) for the Peer Review

Review of the Alaska-wide Sablefish Assessment

CIE reviewers are contracted to complete their independent peer review based on the ToRs. Therefore, the CIE-NOAA Fisheries review and approval process is based on whether the CIE independent reports addressed each of the ToRs.

1) Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
   a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

2) Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
   a) Advise as to whether the model is consistent with and represents the best scientific information available (BSIA) for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

3) Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Reviewers must provide details for the basis of their responses to the review ToRs.

Annex 3. Agenda for the CIE Review of the Alaska-wide Sablefish Assessment

North Pacific Fishery Management Council

NOAA Alaska Fisheries Science Center, Auke Bay Laboratories

17109 Pt. Lena Loop Road, Juneau, AK 99801

June 16-18, 2026

Contact: Chris Lunsford (Chris.Lunsford@noaa.gov)

Tuesday, June 16, 2026

A. Welcome and Introductions

09:00 — 09:15: Welcoming Remarks, Introductions, Logistics Chris Lunsford

09:15 — 09:30: Announcements, Agenda/ToRs Chair

B. Sablefish Management, Biology, Fishery, and Data Inputs (ToR 1)

09:30 — 09:50: Biology, Spatial Dynamics Daniel Goethel

09:50 — 10:10: Sablefish Management/NPFMC FMP Requirements Sara Cleaver

10:10 — 10:30: Fishery-Dependent Data Matt Cheng

10:30 — 10:50: *BREAK*

10:50 — 11:15: Longline Survey Data and Recent Design Changes Kevin Siwicke

11:30 — 12:00: Assessment History, Inputs, and Other Available Data Daniel Goethel

12:00 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Assessment Model Configuration (ToR 1)

14:00 — 14:30: Current Model Development Pathway Daniel Goethel

14:30 — 15:30: General Sablefish Assessment Parametrization Daniel Goethel

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion

Wednesday, June 17, 2026

A. Reconvene

09:00 — 09:15: Announcements, Agenda/ToRs Chair

09:15 — 10:30: Discussion

10:30 — 10:50: *BREAK*

B. Spatially Implicit (Fleets-as-Areas) Model Configuration (ToRs 1)

10:50 — 12:00: Results of Spatial Simulations to Optimize FAA Structure Matt Cheng

12:00 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Comparing Assessment Performance and Diagnostics (ToR 1/2)

14:00 — 14:20: Final Fleets-as-Areas Assessment Configuration Daniel Goethel

14:20 — 15:30: Outputs/Diagnostics for FAA and Panmictic Models Daniel Goethel

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion

Thursday, June 18, 2026

A. Reconvene

09:00 — 09:15: Announcements, Agenda/ToRs Chair

B. Comparing Assessment Performance and Diagnostics (ToR 1/2)

09:15 — 10:30: Discussion

10:30 — 10:50: *BREAK*

10:50 — 11:30: Results of a Spatial Closed Loop Simulation Analysis. Matt Cheng

11:30 — 12:30: Discussion

12:30 — 14:00: *LUNCH BREAK*

C. Spatial Models (ToR 3)

14:00 — 15:00: 3 and 5 Region Spatial Model Configuration/Outputs Matt Cheng

15:00 — 15:30: Discussion

15:30 — 15:50: *BREAK*

15:50 — 17:00: Discussion and Report Writing

ADJOURN

Appendix 3 List of Participants

CIE - Center of Independent Experts

NOAA - National Oceanic and Atmospheric Administration

NMFC – National Marine Fisheries Service

AFSC - Alaska Fisheries Science Center

NPFMC - North Pacific Fishery Management Council

ASMFC - Atlantic States Marine Fisheries Commission

UA – University of Alaska

Klass Hartmann	CIE Reviewer Australia	
Peter Stephenson	CIE Reviewer Australia	
Sam Subbey	CIE Reviewer Norway	
Aaron Berger	Meeting Chair	(NOAA/NMFS NWFSC)
Daniel Goethel	Assessment Presenter	(NOAA/NMFS AFSC)
Matt Cheng	Assessment Presenter	(NOAA/NMFS AFSC; UA)
Chris Lunsford	Meeting Organizer	(NOAA/NMFS AFSC)
Kristen Omori	NOAA/NMFS AFSC	
Pete Hulson	NOAA/NMFS AFSC	
Kevin Siwicke	NOAA/NMFS AFSC	
Sara Cleaver	NPFMC	
Pat Malecha	NOAA/NMFS AFSC	
Melissa Haltuch	NOAA/NMFS AFSC	
Grant Adams	NOAA/NMFS AFSC	
Jim Ianelli	NOAA/NMFS AFSC	
Cindy Tribuzio	NOAA/NMFS AFSC	
Samara Nehemiah	ASMFC	
Beth Matta	NOAA/NMFS AFSC	
John Brogan	NOAA/NMFS AFSC	

Appendix 4. Review Panel Summary Report.

TOR 1. Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.

a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

- Trade-offs of adding flexibility using FAA, but then restricting selectivity to only logistic brings you back some. Keep an open mind on some of the alternative selectivity-by-region scenarios that also performed relatively well. Including some alternative selectivity forms beyond gamma.
- Look at some intermediate complex selectivity parameterizations for the time-varying options (or time/age interaction).
- The FAA approach overcomes the difficulties with spatial/temporal survey design, which additionally addresses the most recent design changes.
- FAA adheres to the management structure and retains applicability to the current management framework/advice.
- FAA is a good compromise between capturing sufficient biological complexity and overparameterization (parsimonious model structure)
- Future improvements are unlikely to arise from additional to the presented time-varying selectivity complexity, alternative composition likelihood formulations, or further fleet partitioning. Extensive sensitivity analyses indicate that these approaches do not resolve the fundamental model tensions.
- The current data do not appear sufficient to support a fully spatial process-based assessment model. Additional spatial complexity would likely exacerbate existing identifiability and stability issues unless accompanied by substantial improvements in data quantity, spatial coverage, and understanding of ecological processes.
- Tagging information represents an underutilized source of information and offers opportunities to improve understanding of movement, survival, and stock connectivity. However, methods for integrating tagging information into assessment models requires further development and validation.

TOR 2. Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.

a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

- FAA is more difficult to interpret (especially selectivity and catchability relative to a one-area interpretation) than a single area model.
- FAA will likely be difficult to communicate to stakeholders.

- The additional parameters required for the FAA model were validated as appropriate given the presented closed loop simulation analyses. It would be beneficial to examine if the results are robust to alternative operating models (states of nature).
- Natural mortality remains difficult to estimate (regardless of single-area or FAA approach). Currently, the estimate of natural mortality is driven by prior, which directly influences population scale and can be confounded with catchability and R0.
- Weakness is that the FAA relies on the survey data for regional dynamics, but the survey does not fully cover the stock footprint, particularly in the AI region. [this weakness remains for other approaches too].

TOR 3. Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

- Investigate Jeffreys priors (i.e., non-informative priors) for natural mortality, which can be helpful when there is a lack of information on what to specify as the prior.
- Investigate the tagging data for estimating M (outside the assessment model), in particular towards achieving an improved prior distribution for use in assessment models.
- Understand the impact of age data quantity on age compositions and, through the application of data weighting, how it influences fits to composition and relative index fits. Will more age data help with age composition fits?
- To make the 3 and/or 5 region spatially-explicit models operational, methods need to be developed to be able to specify spatial reference points. Currently, there is no clear advice on best practices.
- There is a management integration conundrum with the spatial models because the 5-region has a management mismatch and the 3-area has a biological mismatch that would have to be dealt with.
- The fits to the composition data remain an area for model improvement. Investigate whether new information on aging error, the assumed additional survey index CV, selectivity, and other approaches lead to improved fits.

Additional Notes:

- The continued collection of otoliths and aging should at least stay at the current level.
- The survey data and footprint have already been diminished to a level that is significantly impacting model uncertainty, and further reductions would seriously compromise the assessment.
- Extending the current hydrodynamic/environment models (MOM-6 now available) to additionally include the BS and AI area (not just the GOA) would be beneficial to understand dispersal dynamics in these areas.
- While the removal of data (skates) on the survey due to whale depredation is generally formalized on the water, though remains subjective, a formal description of these protocols should be completed.